

A COMPUTATIONAL DESIGN SCIENCE APPROACH FOR INTER-VEHICLE NETWORK COMMUNICATION USING LEO SATELLITE INTERNET

Version 1.04/2025.08.13

Lappeenranta–Lahti University of Technology LUT

Degree Programme in Software Engineering, Master's thesis

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Examiners: Title First name and Last name (optional degree)

Title First name and Last name (optional degree)

Abstract

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LUT School of Engineering Science

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Declarations

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 - b. Explanation of the use of the tool:

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The author, Aleksandar Ivanov, takes full responsibility for the content of this thesis and has reviewed and edited the content generated by the possible use of AI tools.

Table of contents

Abstract	
Symbols and abbreviations	
Declarations	
Lists of figures and tables (optional)	6
1 Introduction	7
1.1 CBRN Defense Field	7
1.2 LEO Satellite Networks	7
1.3 QUIC Protocol	8
1.4 Motivation and Context	8
1.5 Objectives and Scope of This Work	8
2 Theoretical background	11
2.1 CBRN defense systems	11
2.2 Low Earth Orbit (LEO) Satellite Networks	12
2.3 QUIC Protocol	12
3 Third section	13
3.1 Second level subsection	13
3.1.1 First subsubsection	13
3.1.2 Second subsubsection	13
3.2 Subsection	13
4 Conclusions	14
References	15
Appendices	
Appendix 1 Text processing and layout in a thesis	
Appendix 2 References	
Appendix 3 Tables, figures, equations, numbers, symbols, and abbreviations	
Appendix 4 Appendices to the thesis	
Appendix 5 Publishing the thesis	

Appendix 6 Matlab script for example figure

Appendix 7 Bibliography entries in .bib file

List of Figures

- 1 Illustration of Matlab example figure.
- 2 Short description for List of Figures
- 3 Example to define PDF meta-data in latex-system.

List of Tables

- 1 Sensor measurements.

1 Introduction

1.1 CBRN Defense Field

CBRN defense field encompasses the military and civil measures taken to protect against chemical, biological, radiological, and nuclear threats. The field includes a wide range of activities, such as detection, identification, decontamination, and medical countermeasures (Otríasal, 2014). CBRN events can pose a significant physical and psychological hazard to the public and pose environmental and economic impact (Ministry of Health Malaysia, 2025). Fast, coordinated and accurate response in the first critical hour or two is crucial to minimize the impact of such events (Joint Emergency Services Interoperability Programme (JESIP), 2016). Providing accurate information in a timely manner is essential for effective decision-making and response (Ministry of Health Malaysia, 2025).

Coordination between the various agencies and personnel involved in CBRN defense is essential. Numerous software solutions have been developed to facilitate communication and information sharing among CBRN defense teams (Marques, Duarte & Lobo, 2025; Environics Oy, 2026; Bruhn NewTech, 2021; Observis Oy, 2026). A common pattern among these solutions is the near real-time consolidation and visualization of data from multiple sources, such as sensors, first responders, command centers and various governmental databases. However, the effectiveness of these solutions is limited by the communication infrastructure available in the affected area. Additionally, any delays introduced by the communication infrastructure, protocols, and data processing can add up to significant delays in the overall response time. In the case of CBRN events, where every second counts, these delays can have serious consequences.

1.2 LEO Satellite Networks

One of the more recent developments in communication infrastructure is the commercial availability of low earth orbit (LEO) satellite internet. LEO satellite networks have the potential to provide high-speed, low-latency internet access to remote and underserved areas (Duan & Dinavahi, 2021). This technology has the potential to improve the communication abilities of CBRN defense teams by providing a communication channel in remote areas or areas with deliberately disrupted communication infrastructure (Abels, 2024).

One of the leading providers of LEO satellite internet is Starlink, which as of September 2024 has launched 6426 satellites into orbit and has 14 million active subscribers globally, making it the leader by a wide margin in both of these metrics. Starlink subscriptions are widely available across the globe in both personal and business plans (Koosha, Madani &

Ardakani, 2025). The relatively low subscription cost, easy of acquisition, low-latency, high bandwidth and coverage make it a compelling option as the communication infrastructure for CBRN defense teams.

1.3 QUIC Protocol

QUIC is a multiplexed, encrypted and low-latency transport protocol that is designed with the goal of improving transport performance and security of internet applications. QUIC is a cross-layer protocol and operates at the Transport, Security and Application layers of the OSI model (Langley et al., 2017).

According to the designers of the QUIC protocol, two of the main motivators behind its design were reducing the handshake delay of TCP and TLS and Head-of-line blocking delay within TCP. Additionally, TCP is a rigid protocol which cannot be modified to suit the needs of modern applications or specific use cases. QUIC comes out as an answer to these limitations and by measurements from Google it reduces latency of Google searches by 8% on desktop and by 3.6% on mobile. (Langley et al., 2017) The flexibility and low latency characteristics of QUIC makes it particularly suitable for the demanding application of CBRN monitoring and communication over LEO satellite networks.

1.4 Motivation and Context

This thesis aims to address the challenges of near real-time communication for CBRN applications with global coverage by proposing a computational design science approach for inter-vehicle network communication using QUIC as a transport protocol over LEO satellite internet. The proposed approach leverages the advantages of LEO satellite internet, such as low latency and high bandwidth, to enhance the communication capabilities of CBRN defense vehicles. QUIC is used as a transport protocol due to its flexibility for specific applications and low latency. By improving the speed and accuracy of information sharing, the proposed approach has the potential to significantly enhance the effectiveness of CBRN defense operations. The proposed design aims to enable wider capabilities of software systems for CBRN defense, such as global connectivity in remote areas and reduced latency and reliability over traditional network infrastructure (e.g. 3G, 4G, LTE, 5G and geostationary satellite internet) and transport protocol.

1.5 Objectives and Scope of This Work

This thesis adopts the Design Science Research Methodology (DSRM) framework by Peffers et al. (2007) to design, implement, and evaluate a proof-of-concept (PoC) software communication artifact. The primary aim of the artifact is to leverage the QUIC transport protocol

over Low Earth Orbit (LEO) satellite internet for near real-time CBRN communication. Furthermore, Machine Learning (ML) will be used to optimize the performance of the QUIC protocol over LEO satellite networks, with a focus on minimizing latency, improving resilience and prioritizing critical CBRN data payloads. The artifact will be evaluated in a real-world environment using Starlink’s commercial LEO satellite network.

To achieve this aim, the work is divided into the following concrete research objectives:

- **O1 (Artifact Design and Implementation):** Develop a software prototype capable of transmitting and receiving CBRN data payloads between remote vehicular nodes using the QUIC transport protocol with ML optimizations.
- **O2 (Payload Simulation):** Design a standardized, synthetic data schema that mimics real-world CBRN sensor telemetry (e.g., detection alerts, GPS coordinates, and timestamped hazard measurements) without relying on proprietary hardware.
- **O3 (Empirical Evaluation):** Measure the performance of the implemented QUIC artifact over a live commercial LEO satellite network (Starlink) under real-world operating conditions, tracking metrics such as end-to-end latency, handshake duration, packet loss, and connection recovery during satellite handovers.
- **O4 (Comparative Benchmarking):** Compare the performance of the QUIC-based artifact with and without ML optimizations against a traditional baseline protocol stack (i.e., TCP with TLS 1.3) running over the same LEO satellite network.

To guide the empirical evaluation of the artifact, this study formulates two core research questions (RQs) and their corresponding testable hypotheses (H_1 and H_2):

RQ1: How does replacing traditional TCP/TLS transport protocols with standard QUIC affect the latency, connection resilience, and real-time data delivery of CBRN telemetry when routed over commercial LEO satellite networks?

Hypothesis 1 (H_1): Because of its 0-RTT/1-RTT handshake mechanism and elimination of head-of-line blocking, utilizing the standard QUIC protocol over LEO satellite internet significantly reduces end-to-end transmission latency and improves connection recovery times during network dropouts compared to a standard TCP/TLS baseline.

RQ2: To what extent does applying Machine Learning (ML) optimizations to the QUIC protocol further enhance packet prioritization and resilience during predictable LEO network degradations, such as satellite handovers?

Hypothesis 2 (H_2): An ML-optimized QUIC implementation will yield a statistically significant reduction in latency for high-priority CBRN payloads during satellite handovers compared to a non-optimized QUIC baseline, by dynamically adapting transport parameters to the fluctuating LEO network state.

Delimitations and Scope: To maintain scientific rigor and feasibility within the timeframe of a Master’s thesis, explicit boundaries are set regarding the scope of this study:

- **Network Infrastructure:** The empirical testing is strictly limited to the Starlink commercial LEO satellite constellation. Other LEO constellations (such as OneWeb or Amazon Kuiper), geostationary (GEO) satellites, and terrestrial cellular networks (e.g. 3G, 4G, LTE, 5G) are excluded from the experimental setup.
- **Hardware and Sensors:** Physical CBRN detection hardware, chemical analyzers, and vehicular sensor integration are out of scope. The study relies entirely on software-simulated data payloads that accurately replicate the data volume and frequency of standard CBRN defense devices.
- **C4I Integration and UI:** While the artifact demonstrates data transport capabilities, developing a full Command, Control, Communications, Computers, and Intelligence (C4I) situational awareness dashboard or graphical user interface (GUI) is out of scope. The focus remains strictly on the networking and transport layers.
- **Adversarial Security Testing:** Although QUIC inherently enforces TLS 1.3 encryption, active electronic warfare testing—such as intentional radio frequency (RF) jamming, satellite spoofing, or deep cryptographic penetration testing—is excluded from this work.

2 Theoretical background

The following sections provide a theoretical background of the various technologies, protocols, standards and scientific approaches relevant to this thesis work.

2.1 CBRN defense systems

CBRN defense systems are systems which are designed to aid first responders with their strategic decision making, standard operating procedures and situational awareness before, during and after the occurrence of CBRN incidents. CBRN defense refers to the defense against chemical, biological, radiological and nuclear threats.

CBRN defense systems are used by militaries, first responders and dedicated CBRN defense units. These systems are commissioned by governments in accordance with CBRN preparedness goals and strategies. Such systems typically consist of hardware and software components which work together to collect, process and present information related to CBRN incidents. The hardware can include CBRN detectors, sensors, communication devices, personal computers, laboratory equipment and various mechanical and electrical automation systems. The exact hardware depends on the purpose of the system and the environment it is designed to operate in. (Policastro, 2008; Thompson & Gahlaut, 2015 - 2015)

The software components are used to collect and process information from the hardware components and provide it to the users of the system. A common function of these types of systems is to provide near real-time measurements and alerts from collected sensors and other information sources. Furthermore, the information is typically presented to users using a map-based interface, where measurements, detected hazards, personnel, equipment and other relevant information are shown in reference to their geographical location. This allows information from multiple sources to be presented to users in a way that improves their situational awareness.

CBRN defense systems typically record all available information within the system by design, which enables users to perform post event analysis of CBRN incidents. Reviewing and cross referencing plots of recorded measurements and locations, generating and managing reports, overall incident review and incident documentation are common activities. These features can be typically designed in accordance with organizational standard operating procedures and regulations, reducing the need for manual documentation and reporting and increasing the accuracy of results from such procedures. Furthermore, the ability to perform post analysis of CBRN incidents can be used to improve organizational performance and preparedness for future incidents.

These systems need to provide information with low delay and sufficient accuracy for decision making. Reliability, promptness and availability are important features of these systems, as they may operate during situations in which the communication infrastructure or individual sub-components are unavailable or degraded. Each sub-component of the software system can introduce delays and inaccuracies, act as a bottleneck or a point of failure. Therefore, every layer of the system and interfaces between sub-components need to be designed with rigorous performance and reliability requirements.

2.2 Low Earth Orbit (LEO) Satellite Networks

Low Earth Orbit (LEO) satellites are satellites that orbit the earth at a speed higher than that of earth's rotation. As a result of that higher speed, LEO satellites appear to continuously circulate around the earth. In order to provide global coverage a constellation of LEO satellites is required. (Sun, 2014)

2.3 QUIC Protocol

3 Third section

3.1 Second level subsection

3.1.1 First subsubsection

3.1.2 Second subsubsection

3.2 Subsection

4 Conclusions

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Appendix 1 Text processing and layout in a thesis

Good text processing skills make writing your final thesis and using this template easier. Therefore, you should ensure you have sufficient basic skills to edit long documents with $\text{\LaTeX}$ before you start. This involves applying $\text{\LaTeX}$ code structures, understanding automatic referencing and knowing how to divide your text into sections.

The essential thing is to understand the following basics of $\text{\LaTeX}$:

- $\text{\LaTeX}$ needs a compiler. Compiler can be stand-alone in the computer or cloud-based, such as [Overleaf](#)
- $\text{\LaTeX}$ will remove all the unnecessary spaces and linebreaks. When you want to start a new paragraph, make one empty line into your code, $\text{\LaTeX}$ will start the paragraph.
- Do not enter any numbering manually. $\text{\LaTeX}$ has efficient automatized tools for this. Section, page, figure, and table numbering are automatically applied in this template. They keep the numbers in the right order even if you modify, add or remove content.
- Do not add hyphenation at the end of a line manually. $\text{\LaTeX}$ automatic hyphenation tool can be used in this template. It should work with both English and Finnish languages.

Line spacing, font, margins, alignment, page numbering and headings

Official layout guidelines state that the line spacing should be 1.5, except for the abstract and possible direct citations, where the spacing is 1. You can choose from two fonts: Times New Roman (12 pt) or Arial (11 pt). This template has been written in Adobe Times Roman 12 pt.

Leave the following margins:

- top and left 35 mm.
- bottom and right 20 mm.

The page number of the title page is 1, but the page numbering should not be visible before the first page of the table of contents. Place the page numbers at the top, either centred or in the right-hand corner. Page numbering ends on the final page of the reference list: appendices do not have page numbers unless the appendix is multiple pages. For a single-page appendix, use the command `\thispagestyle{empty}` to remove the page number.

Always use section styles in your thesis (`\section{}`, `\subsection{}`, `\subsubsection{}`).

Always place section headings (`\section{}`) on a new page. Apply `\clearpage` command before the section heading starts the section from the new page. If you add large figures or tables, remember to check that the empty space after the figure or image covers no more than 20% of the page.

The heading numbering starts from the introduction and continues consecutively. Do not place a period after the heading number. Do not number the heading of the reference list at the end of the thesis.

The thesis should include no more than three heading levels, and the headings should progress logically. If you need even more detailed subheadings, do not number them; leave them out of the table of contents. Concise headings that describe the text sufficiently are the best. You can use question marks or exclamation marks but do not add a period if the heading is a regular sentence.

A heading cannot follow a heading. Always write something between them. For instance, there must be text between headings 1 and 1.1 with two or more sentences.

Lists are a good way to express things clearly. Use the same type of bullet or symbol in lists throughout your thesis. A section should never end in a list. There should always be two or three sentences after a list.

Appendix 2 References

The text must include references to the sources you use. LUT University applies the Harvard referencing style, also called the author-date style, with in-text referencing and a detailed reference list at the end.

The purpose of a reference list is to provide sufficient information on a source used in the study, allowing the reader to consult the original source for further information. The reference enables the reader to find detailed information on the source easily in the list of references. You should refer to the original and most recent sources. If no new studies have been published on the topic in question, also older ones may be used.

Referring to a source means that you explain the contents of the source material in your own words. Direct citations, on the other hand, are placed inside quotes “Quote”. Plagiarism or using another person’s original material without appropriate referencing is not allowed.

Referencing technique

In the Harvard system, the citation is placed in parentheses directly in the text to indicate the passage that has been cited from another source. Place the citation before the period that ends the sentence when it refers only to the sentence in question (Geldart, 1985). If you are referring to more than that one sentence, introduce the source you are summarizing or paraphrasing at the beginning of the paragraph. Then, refer back to the source when needed to ensure your reader understands you are still using the same source. In-text references, specifying page numbers, are not used unless a specific part of the text is directly referred to or it is a direct quotation. Please note that the author does not always have to be an individual but can also be, for example, an organization. If the publication’s author does not appear in the source, the name of the publication is referred to instead of the author’s name.

Author’s name cited in the text (direct)

Direct citation is used when the author’s name(s) is a part of the sentence structure, with the year in parentheses. The following is obtained by using `\citet{key}` command.

Allen & Tildesley (1987) claims that ...

More than three authors for a work

Chowdhury et al. (2017) in their work ...

On some special occasions, all the authors (more than three) can be cited, but this form should be considered only at the *special* case by using `\citet*{key}` command

Bizon, Shattuck, Swift, McCormick & Swinney (1998) showed ...

Author's name not cited in the text (indirect)

Indirect citation is used when the author's name(s) is not mentioned directly in the sentence structure. The indirect citation will appear at the relevant point in the sentence or at the end of the sentence in parentheses. The following is obtained by using `\citep{key}` command.

Earlier research claims that ... (Savage, 1992).

Two authors with postnote

At the theoretical approaches of ... (Allen & Tildesley, 1987, p. 102).

More than three authors for a work

..., (Veleva et al., 2015).

Combined citation with three entries using the command `\citep{key1,key2,key3}`

According to previous studies, the presented conclusions are consistent (Miller, 1997; Dalton, Petri & Pontuale, 2005; Allen & Tildesley, 1987).

In many situations, it is justified to combine the content of several sources, and then it is recommended to use content-specific indirect references. For example, in the following way:

“In literature we can find challenges relating to a wide variety of issues including environmental sustainability (Dagiliute & Liobikiene, 2015), business needs (Veleva et al., 2015), food production (Chowdhury et al., 2017), urban development (Joss, 2010; Swapan, 2016; Freeman, McMahon & Godfrey, 2017; Waite et al., 2017), and education (Zidansek, 2007; Biberhofer & Rammel, 2017).” (Talikka, 2018)

A wide variety of sources can be used in the text. In addition to the above, Master's theses (Hyvärinen, 2019), websites (NASA, 2015), interviews (Kottarainen, 2007, interview 6.6.2007), technical reports (Laine, Leppänen & Nurmi, 2009), booklets (Swetla, 2015), manuals (R Core Team, 2018), proceedings (Stepney & Verlan, 2018), standards (ISO 9004, 2000), patents (Virtamo & Aalto, 1996) or AI-language models (OpenAI, 2023) can be used.

All the bibliography information is given in the References. If you refer several works published by the same author in the same year, add lowercase letters (a, b, c . . .) after the publication year to distinguish the sources. Use the same alphabetical notation also in the list of references. There are several referencing and citation styles. It is essential to use the same style throughout the thesis consistently. Examples and detailed instructions on referencing:

- [LUT Academic Library's instructions](#) on how to cite electronic documents
- [Aalto University citation guide](#)
- [Harvard referencing](#), University of Sheffield

Appendix 3 Tables, figures, equations, numbers, symbols, and abbreviations

It is a good idea to illustrate your text with figures and tables. Figures and tables must have captions and consecutive numbering. The captions of tables are placed above the table, and those of figures are below the figure. Refer to the figures and tables in the text body, preferably before you introduce them, and align them with the text body. In many theses, the alignment of figures and tables has been centered and without wrapping text around them, but the author can select the formatting options as long as they are coherent in the whole work. $\text{\LaTeX}$ tries to place all floating objects (figures and tables) at the top of the page or in the best place and to avoid multiple floating objects on the same page. Because of this, for several floating objects in series, their position can be shifted several pages forward. It is also possible for floating objects to move to the next section. Because of this, several consecutive floating objects should be avoided, and enough text should be added between the floating objects. In this template `\usepackage[section]{placeins}` setting is used to force floating objects to stay in the appearing section. You can also use `\FloatBarrier` command to set a position limit for floating objects.

Remember to add alt text (alternative text) to your figures and tables to ensure accessibility. Alt text is read by a designated reader and can be viewed even when the image cannot be displayed on the page. In $\text{\LaTeX}$ the alt text has to be set manually and you should make sure it describes the object sufficiently and understandably. You can add the alt text by using the command `\pdftooltip{Object}{Alt-text}`.

Tables

Give your table number and caption, and refer to them in the text. Place the caption above the table, name its columns, and mention the units applied, as in Table 1 below. Avoid empty columns or rows. The recommended font size is 10.

Table 1: Sensor measurements.

Voltage U [V]	Pressure p [Pa]
0.984	0
2.252	150
2.772	300
3.181	450
3.615	600
3.817	750
4.088	900

Figures, charts, graphic elements

Images help illustrate your text. The text should contain a reference to every figure. Number your figures and place a caption underneath – not inside the figure.

You should use a software program like Excel or Matlab to draw charts. Charts should be clear and easy to understand. Use a white background. A background grid is allowed if it does not make the figure difficult to interpret. Variables and measurement points should be clearly visible. Name the axes and their units. An example script to produce a Figure 1 with Matlab is presented in Appendix 6. To maintain full accessibility of the document, you should use embedded fonts in the vector figures. The example script will not produce a figure with embedded fonts, and the jpg or png formats would be better with respect to document accessibility. However, the figure quality is reduced.

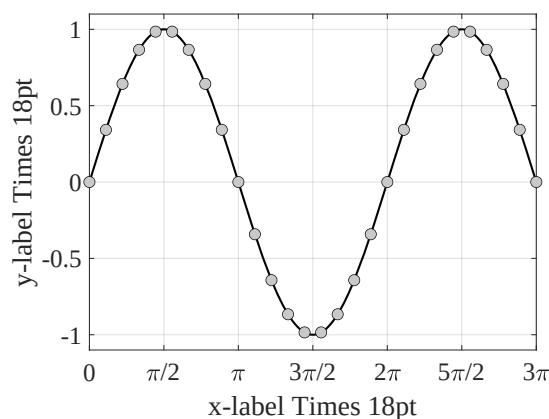


Figure 1: Illustration of Matlab example figure.

Create as many of the figures yourself as you can. Use the same font as in the text body and equations. Try to maintain the font size in the figure as close as possible to the main text font size. If you use images created by someone else, remember to cite them correctly. Remember that images are copyrighted works, the use of which must always be authorized by the author. Captions need to be in the same language as the text body.

Figure 2 shows a gas fermenter in a laboratory configuration. The image is positioned at the beginning of the page, and with the command `\FloatBarrier`, the following text is forced to appear after the image.

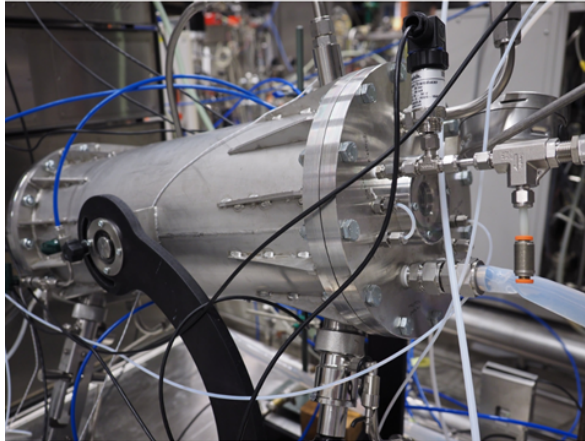


Figure 2: Gas fermentor (VTT 2020, LUT image bank).

Do not end a paragraph in a figure or table. Add text underneath, such as comments on the figure. Large figures, tables, long equations, and other supporting material can be appended.

Numbers, symbols and equations

Numbers in the text are usually approximations. Their accuracy depends on the observational error. Include only significant figures in the results. Interim results should include at least two figures more to avoid round-off errors. Present large and small figures in powers of ten 10^n , where n should preferably be divisible by three.

Equations and other mathematical expressions must consist of standardised symbols if one exists. You may use other symbols only if there are no applicable standardized or established ones. Variables and constants are denoted using *slanted style*, vectors are denoted using **bold regular style**, and abbreviations and dimensionless numbers are denoted using regular style. See the SI-unit [style guide](#).

Explain the symbols in an equation when you use them for the first time. Write each equation clearly on its own line and indent it. Number your equations consecutively or by paragraphs so that the number is in parentheses on the right side of the equation and aligned to the right. You can refer to an equation only after you have presented it, with certain exceptions, such as if the object you are referring to is far ahead. As a simple example, the equation of state of an ideal gas can be used

$$pv = RT, \tag{1}$$

where p is pressure [Pa], v is specific volume [m^3/kg], R is the gas constant [$\text{J}/(\text{kgK})$] and T is temperature [K].

The SI-unit style should be followed. Here are some common guidelines for using variables, constants, and units correctly:

- Write scalars in italics and vectors in bold, not in italics. Don't write dimensionless quantities in italics.
- Don't use * multiplier in your equations. Present the variables to be multiplied in the equations in consecutive order, placing the numerical scalar value first, without any multiplier sign. Exceptions: for 1.452×10^6 or when you place numerical values into the equations in the example calculations, use \times command for a multiplier. In the latter case, remember to show the units as well.
- Use only single-character variables to avoid mistakes with previous point. You may use subscripts to separate variables.
- Write subscripts upright unless there is a need to italicise them (they are variables). Write abbreviated subscripts and numerals e.g., as follows: $\Delta\sigma_w$, σ_1 , or $\sigma_{\min}$. For instance, in the summation $\sum_i^\infty x_i$, the subscript i must be italicised because it represents a variable.
- If you wish to express a change in, e.g. pressure Δp , write Δ in a regular font. In some cases, Δ may also be variable and should be italicised. $\pi = 3.14159$ is the ratio of a circle's circumference to its diameter. π may be the pressure ratio.
- Do not italicise mathematical operators such as $\sin(x)$, $\cos(x)$, $\log(y)$, or $\ln(y)$.
- Distinguish absolute values as follows: "variable=_value_unit", with the exception of a percentage sign after a numeral, e.g. $a = 5.2 \text{ mm}$, $\gamma = 97.7\%$.
- Use a decimal point "." in accordance with international standards. In contrast, a decimal comma is used in theses written in Finnish. This also applies to figures and tables.

List of symbols and abbreviations

List symbols and abbreviations and their definitions that are not common knowledge separately on their own page before the table of contents. Divide them into groups: Latin characters, Greek characters, subscripts, superscripts, abbreviations, and finally, dimensionless numbers. Give the page the heading Symbols if there are no abbreviations or Abbreviations if there are no symbols.

When you use a symbol or abbreviation in the text body for the first time, introduce it to the reader, for example, as follows: "The concept design for manufacturing and assembly (DFMA) is...". After this, you can use only the abbreviation, and the reader can verify its

meaning from the abbreviation list. Do not add concepts to the list of symbols and abbreviations you do not mention in your text. Don't include standard, well-known elements or species (e.g., oxygen O₂ or carbon dioxide CO₂) definitions in the abbreviations.

Appendix 4 Appendices to the thesis

Appendices may include, e.g., interview questions, survey forms, or other content relevant to the work but not necessary to include in the text body. All appendices are public and should not contain any confidential information.

In your text body, refer to the appendices by adding their title in parentheses (Appendix 1) where relevant. Give all appendices a title based on their content and list them in the table of contents in the order in which they are referred to in the thesis.

Single-page appendices do not require page numbering. Multiple-page appendices do.

Appendix 5 Publishing the thesis

LUT's degree regulations state that Bachelor's and Master's theses are public documents. They are published in the LUTPub repository, and related instructions are available on the [library website](#).

Together with the first examiner, make sure that the commissioner of your thesis is aware of the publicity requirements from the very beginning of the discussions. The student writing the thesis and the thesis commissioner are responsible for ensuring that there is no confidential material in the thesis. The background material of the thesis may contain confidential content. The background material is not public.

All theses in LUTPub must fulfill accessibility requirements. Your text must be as legible as possible to readers. Remember:

- to add PDF meta-data at the beginning of main.tex according to Figure 3 with your own information.
- to use $\LaTeX$ commands for headings,
- embed hyperlinks into your text or a description of the linked content; do not include URL addresses,
- add/check the alt text of your figures and tables describing briefly the main content in writing.
- to refer to each figure and table in the text, and verbally describe in the text the key content presented in the figures and tables.

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Give first the meta-data for the PDF/A
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% IMPORTANT!!!
%% Mandatory data, update with your own data
%% This data will appear in the pdf file properties %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
\begin{filecontents*}[overwrite]{\jobname.xmpdata}
\Title{Title of the work}
\Author{Author_firstname Author_lastname}
\Language{en-GB}
\Subject{The abstract or short description.}
\Keywords{template\sep thesis\sep LaTeX}
\end{filecontents*}
```

Figure 3: Example to define PDF meta-data in latex-system.

At the moment (09/2023), $\LaTeX$ text system and its add-ons (packages) can only produce pdf files according to accessibility criteria to a limited extent. Using the $\LaTeX$ system, it is not possible to produce a final product that meets all accessibility criteria and works universally.

Due to deficiencies, the automatic read-out of the PDF file may have problems, especially with images, tables, and equations. For this reason, it is especially important that the contents of floating objects (images, tables, and equations) are described as clearly as possible in the body text.

Appendix 6 Matlab script for example figure

```
%% This script will generate an example figure and it will export the
    figure in jpg, png and pdf format
% Jouni Ritvanen, 2023

close all
clear

fs=18;                % Font size
fc=[0.79 0.79 0.79]; % marker face color (light grey)

x=0:pi/9:3*pi;      % creating x-vector

figure % create figure
fplot(@sin,[0,3*pi],'k','LineWidth',1.5) % plotting sin(x) function
hold on % this allows to add next plot into the figure
grid on % adding grid
plot(x,sin(x),'o',... % plotting discrete sin(x) points
     'MarkerSize',8,... % setting marker size to 8
     'MarkerEdgeColor','k',... % setting marker edge color to black
     'MarkerFaceColor',fc) % setting marker face color to fc

set(gca,'fontname','times') % setting font to Times
set(gca,'fontsize',fs) % setting font size to fs
ylim([-1.1 1.1]) % setting y-axes limits
xlabel('x-label Times 18pt') % setting x-axes label
ylabel('y-label Times 18pt') % setting y-axes label
xticks(0:pi/2:3*pi) % setting x-ticks every pi/2
% setting x-ticks labels
xticklabels({'0','\pi/2','\pi','3\pi/2','2\pi','5\pi/2','3\pi'})
%% this section will adjust the paper size to correspond the figure size
% i.e. the output will be the size of Matlab figure, not e.g. A4
pps=get(gcf,'paperposition'); % gets figure size and position
set(gcf,'papersize',pps(3:4)); % sets paper size to correspond figure
    size
set(gcf,'paperposition',[0 0 pps(3:4)]); % sets figure to fill paper
%% this section will save the figure in jpeg and pdf formats
print -djpeg -r600 example_name % output will in resolution of 600
print -dpng -r600 example_name % png resoluutiolla 600
print -dpdf example_name % vector pdf output
```


Appendix 7 Bibliography entries in .bib file

%% Reference examples

% Article , two authors

```
@article{DAGILIUTE,
  title      = {University contributions to environmental
                sustainability: challenges and opportunities from the Lithuanian case
                },
  journal    = {Journal of Cleaner Production},
  volume     = {108},
  pages      = {891–899},
  year       = {2015},
  doi        = {10.1016/j.jclepro.2015.07.015},
  author     = {Renata Dagiliute and Genovaite Liobikiene}}
```

% Article , five authors

```
@article{VELEVA,
  title      = {Understanding and addressing business needs and
                sustainability challenges: lessons from Devens eco-industrial park},
  journal    = {Journal of Cleaner Production},
  volume     = {87},
  pages      = {375–384},
  year       = {2015},
  doi        = {10.1016/j.jclepro.2014.09.014},
  author     = {Vesela Veleva and Svetlana Todorova and Peter Lowitt
                and Neil Angus and Dona Neely}}
```

% Article , four authors

```
@article{CHOWDHURY,
  title      = {Key sustainability challenges for the global
                phosphorus resource , their implications for global food security , and
                options for mitigation},
  journal    = {Journal of Cleaner Production},
  volume     = {140},
  pages      = {945–963},
  year       = {2017},
  doi        = {10.1016/j.jclepro.2016.07.012},
  author     = {Rubel Biswas Chowdhury and Graham A. Moore and Anthony
                J. Weatherley and Meenakshi Arora}}
```

% Article , one author

```
@article{swapan,
  author     = {Abu Yousuf Swapan},
  journal    = {Journal of Urban and environmental Engineering},
  number     = {2},
  pages      = {221–232},
  title      = {Holistic strategy for urban design: {A} microclimatic
                concern to sustainability},
```

```

volume          = {10},
doi             = {10.4090/juee.2016.v10n2.221232},
year           = {2016}}
% Article , three authors
@article{freeman ,
author          = {Rachel Freeman and Chris McMahon and Patrick Godfrey},
title          = {An exploration of the potential for re-distributed
    manufacturing to contribute to a sustainable , resilient city},
journal        = {International Journal of Sustainable Engineering},
volume         = {10},
number         = {4-5},
pages          = {260-271},
year           = {2017},
doi            = {10.1080/19397038.2017.1318969}}
% Article , six authors
@article{WAITE,
title          = {Global trends in urban electricity demands for cooling
    and heating},
journal        = {Energy},
volume         = {127},
pages          = {786-802},
year           = {2017},
doi            = {10.1016/j.energy.2017.03.095},
author         = {Michael Waite and Elliot Cohen and Henri Torbey and
    Michael Piccirilli and Yu Tian and Vijay Modi}}
% Article , one author
@article{ZIDANSEK,
title          = {Sustainable development and happiness in nations},
journal        = {Energy},
volume         = {32},
number         = {6},
pages          = {891-897},
year           = {2007},
doi            = {10.1016/j.energy.2006.09.016},
author         = {Aleksander Zidansek}}
% Article , two authors
@article{Biberhofer ,
title          = {Transdisciplinary learning and teaching as answers to
    urban sustainability challenges},
journal        = {International Journal of Sustainability in Higher
    Education},
volume         = {18},
number         = {1},
pages          = {63-83},
year           = {2017},
doi            = {10.1108/IJSHE-04-2015-0078},

```

```

author          = {Petra Biberhofer and Christian Rammel}}
% Article , five authors
@ARTICLE{ Biz98,
author          = {C. Bizon and M. D. Shattuck and J. B. Swift and W. D.
                  McCormick and H. L. Swinney},
title           = {Patterns in 3{D} Vertically Oscillated Granular Layers
                  : Simulation and Experiment},
journal         = {Physical Review Letters},
year            = {1998},
volume          = {80},
pages           = {57--60},
number          = {1},
doi             = {10.1103/PhysRevLett.80.57}}
% Book
@BOOK{ All87,
title           = {Computer Simulation of Liquids},
publisher       = {Oxford University Press Inc.},
year            = {1987},
author          = {M. P. Allen and D. J. Tildesley},
address         = {New York},
ISBN            = {0-19-855645-4}}
% In book
@INBOOK{ CitekeyInbook2,
author          = {D. Geldart},
title           = {Elutriation},
editor          = {J. F. Davidson and R. Clift and D. Harrison},
booktitle       = {Fluidization},
edition         = {2nd edn.},
year            = {1985},
publisher       = {Academic Press},
address         = {London},
ISBN            = {0-12-205552-7},
pages           = {383--412},
chapter         = {11}}
% On book
@INBOOK{ Joss ,
author          = {S. Joss},
title           = {Eco-cities: a global survey 2009},
editor          = {C.A. Brebbia and S. Hernandez and E. Tiezzi},
booktitle       = {The Sustainable City VI: Urban Regeneration and
                  Sustainability },
year            = {2010},
publisher       = {WIT Press},
address         = {Southampton},
ISBN            = {978-1-84564-432-1},
pages           = {239--250},

```

```

chapter          = {4}}
% In collection
@INCOLLECTION{Sav92,
author           = {S. B. Savage},
title            = {Numerical simulation of Couette flow of granular
materials; spatiotemporal coherence and 1/f noise},
booktitle        = {Physics of Granular Media},
publisher        = {Nova Science Publishers},
year             = {1992},
editor           = {D. Bideau and J. A. Dodds},
series           = {Les Houches Series}}
% In proceedings
@INPROCEEDINGS{Dal05,
author           = {F. Dalton and A. Petri and G. Pontuale},
title            = {Stress fluctuations and the solid/fluid transition in
a sheared granular bed},
booktitle        = {Powders and Grains},
year             = {2005},
editor           = {R. García-Rojo and H.J. Herrmann and S. McNamara},
address          = {Stuttgart, Germany, Jun. 18–22, 2005},
pages            = {353–355}}
% In proceedings
@INPROCEEDINGS{Gol98,
author           = {I. Goldhirsch},
title            = {Kinetics and Dynamics of Rapid Granular Flows},
booktitle        = {Physics of Dry Granular Media},
year             = {1998},
editor           = {H. J. Herrmann and J. P. Hovi and S. Luding},
volume           = {350},
series           = {NATO ASI Series E},
pages            = {371–400},
publisher        = {Kluwer Academic Publishers}}
% PhD thesis
@PHDTHESIS{Mil97,
author           = {B. J. Miller},
title            = {Stress fluctuations in continuously sheared granular
media},
school           = {Duke University},
address          = {Durham, NC},
year             = {1997}}
% DSc thesis
@PHDTHESIS{Talikka,
author           = {Marja Talikka},
title            = {Recognizing required changes to higher education
engineering programs' information literacy education as a consequence
of research problems becoming more complex},

```

```

school      = {Lappeenranta University of Technology},
address     = {Lappeenranta , Finland},
type        = {Doctoral thesis},
year        = {2018},
URL         = {https://urn.fi/URN:ISBN:978-952-335-238-4}}

% Master's thesis
@THESIS{Hyv19,
author      = {Joonas Hyvärinen},
title       = {Techno-economic evaluation of carbon capture
               technologies integrated to flexible renewable energy system},
school      = {Lappeenranta-Lahti University of Technology LUT},
address     = {Lappeenranta , Finland},
type        = {Master's thesis},
year        = {2019},
URL         = {https://urn.fi/URN:NBN:fi-fe2019100731528}}

% Website
@MISC{CitekeyMisc,
title       = {Pluto: The 'Other' Red Planet},
author      = {NASA},
howpublished = {[ Website]},
URL         = {https://www.nasa.gov/nh/pluto-the-other-red-planet},
year        = {2015},
note        = {[ Cited: 2023-03-06]}}

% Technical report
@TECHREPORT{Lai09,
author      = {J. Laine and T. Leppänen and O. Nurmi},
title       = {The influence of special effects on the perception of
               printed products},
institution = {VTT Technical Research Center of Finland},
year        = {2009},
type        = {Research report},
series      = {VTT-R-04966-09},
address     = {Espoo}}

% Interview
@MISC{CitekeyMisc2,
type        = {[ Phone interview] 6.6.2007},
author      = {Brian Kottarainen},
year        = {2007},
note        = {Interviewer Outi Opiskelija , the recording is in the
               author's possession}}

%k Booklet
@booklet{CitekeyBooklet,
title       = {Canoe tours in {S}weden},
author      = {Maria Swetla},
howpublished = {Distributed at the Stockholm Tourist Office},
year        = {2015}}

```

% Manual

```
@manual{CitekeyManual,
  title      = {{R}: A Language and Environment for Statistical
               Computing},
  author      = {{R Core Team}},
  organization = {R Foundation for Statistical Computing},
  address     = {Vienna, Austria},
  year       = {2018}}
```

% Proceedings

```
@proceedings{CitekeyProceedings,
  editor      = {Susan Stepney and Sergey Verlan},
  title       = {Proceedings of the 17th International Conference on
               Computation and Natural Computation, Fontainebleau, France},
  series      = {Lecture Notes in Computer Science},
  volume      = {10867},
  publisher   = {Springer},
  address     = {Cham, Switzerland},
  year       = {2018}}
```

% Standard

```
@TECHREPORT{standardi,
  author      = {{ISO 9004}},
  title       = {Quality magement systems -- Guidelines for performance
               improvements},
  institution = {International Organization for Standardization},
  year       = {2000},
  type       = {Standard},
  address     = {Geneve, Switzerland}}
```

% Patent

```
@MANUAL{US5503249,
  type      = {Patent, US-5503249},
  author     = {J. Virtamo and S. Aalto},
  organization = {Kone Elevator, SW Baar},
  title      = {Procedure for controlling an elevator group},
  year      = {1996}}
```

% AI language model

```
@MISC{ChatGPT,
  type      = {[Large language model]},
  title     = {ChatGPT, Mar 14 version},
  author    = {OpenAI},
  URL      = {https://chat.openai.com/chat},
  year     = {2023}}
```

```
@article{Otrisal2014,
```

```

author = {Pavel Ot\v{r}\'\{i}sal},
title = {The Concept of {CBRN} Defence and Possibilities of its
        Application in the {Czech Armed Forces}},
journal = {Journal of Defense Resources Management},
year = {2014},
volume = {4},
number = {2},
pages = {1--4},
url = {https://www.longdom.org/open-access/the-concept-of-cbrn-
        defence-and-possibilities-of-its-application-in-the-czech-armed-
        forces-21873.html}
}

```

```

@manual{JESIP2016,
  title = {Responding to a {CBRN(e)} event: joint operating
           principles for the emergency services},
  author = {{Joint Emergency Services Interoperability Programme (
            JESIP)}}},
  organization = {HM Government},
  year = {2016},
  note = {Available at: \url{https://www.gov.uk/government/
           publications/counter-terrorism-strategy-contest} (Accessed: 13
           November 2019)},
  address = {United Kingdom}
}

```

```

@manual{MOHMalaysia2025,
  title = {Chemical, Biological, Radiological, Nuclear, and
           Explosives Management Guidelines},
  author = {{Ministry of Health Malaysia}},
  organization = {Disaster, Outbreak, Crisis, and Emergency Management
                  Sector, Preparedness, Surveillance, and Response Section, Disease
                  Control Division},
  year = {2025},
  address = {Putrajaya, Malaysia},
  isbn = {978-967-25827-9-3},
  edition = {First Printing}
}

```

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@inproceedings{Marques2025,
  author = {Mario Monteiro Marques and Filipe Duarte and Victor Lobo},
  title = {{CBRN} Emergency Management System},
  booktitle = {Developments and Advances in Defense and Security},
  year = {2025},
  pages = {319--330},
  publisher = {Springer},

```

```

note      = {Chapter 26. First Online: 31 March 2025}
}

```

```

@misc{EnvironicsEnviScreenX,
  author    = {{Environics Oy}},
  title     = {{EnviScreenX} {CBRN} System Software},
  howpublished = {Official Product Documentation},
  year      = {2026},
  url       = {https://www.environics.fi/cbrn-products/enviscreenx/},
  note      = {Accessed: 4 July 2026},
  address   = {Mikkeli, Finland}
}

```

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@manual{BruhnNewTechFrontline,
  title     = {{CBRNe} {Frontline}: Technical Product Specification
    and Data Sheet},
  author    = {{Bruhn NewTech}},
  organization = {Distributed by BMD S.p.A. Defense and Security Systems
    },
  year      = {2021},
  url       = {https://www.bmdspa.it/docs/511/bruhn-newtech-cbrne-
    frontline.pdf},
  note      = {Accessed: 4 July 2026}
}

```

```

@misc{ObservisObsasLink,
  author    = {{Observis Oy}},
  title     = {{ObsAS LINK} --- Integrated {CBRN} Sensor Networking
    and Communications Framework},
  howpublished = {Official Technical Product Overview},
  year      = {2026},
  url       = {https://www.observis.fi/en/obsas-link/},
  note      = {Accessed: 4 July 2026},
  address   = {Mikkeli, Finland}
}

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@book{Sun2014,
  author    = {Zhili Sun},
  title     = {Satellite Networking: Principles and Protocols},
  publisher = {John Wiley & Sons, Incorporated},
  year      = {2014},
  address   = {Chichester, UK},
  url       = {https://ebookcentral.proquest.com/lib/lut/detail.action?
    docID=7104144},
  note      = {Accessed via ProQuest Ebook Central for LUT University}
}

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@article{DuanTong2021SSNC,
keywords = {Starlink ; Aerospace electronics ; co-simulation ;
Communication ; Communication network ; Communications equipment ;
Communications systems ; cyber-physical power system ; Cyber-
physical systems ; Delays ; Extraterrestrial measurements ; free
space ; Infrastructure ; Internet ; Internet access ; Phasor
measurement units ; power system simulation ; Power systems ;
Satellite constellations ; Satellites ; space network ; Space
vehicles},
language = {eng},
number = {4},
pages = {3673-3675},
publisher = {IEEE},
title = {Starlink Space Network-Enhanced Cyber-Physical Power System},
volume = {12},
year = {2021},
abstract = {The information and communication technology (ICT) is
increasingly involved in the measurement, operation, control and
protection of the cyber-physical power system (CPPS). As one of the
most promising communication infrastructure projects, Starlink is a
satellite Internet constellation being constructed by SpaceX for
providing global Internet access, which can benefit the network
service supply for the communication-enabled power system equipment in
locations where the network access is unreliable, expensive, or
completely unavailable. In this letter, the future applications of the
Starlink space network in CPPSs are explored: the communication
infrastructure and transmission parameters are discussed, and the
corresponding test case was emulated in real-time on the heterogeneous
co-emulation platform to validate the proposed concept of space
network enhanced cyber-physical power system.},
author = {Duan, Tong and Dinavahi, Venkata},
address = {Piscataway},
copyright = {Copyright The Institute of Electrical and Electronics
Engineers, Inc. (IEEE) 2021},
issn = {1949-3053},
journal = {IEEE transactions on smart grid},
}

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@article{AbelsJoscha2024Piig,
keywords = {Business ; Capital ; Companies ; Cooperation ; Decision
making ; Geopolitics ; Globalization ; Independence ;
Infrastructure ; International cooperation ; Multinational
corporations ; Ownership ; Politics ; Transnationalism},
language = {eng},
number = {4},

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pages = {842–866},
publisher = {SAGE Publications},
title = {Private infrastructure in geopolitical conflicts: the case of
        Starlink and the war in Ukraine},
volume = {30},
year = {2024},
author = {Abels, Joscha},
address = {London, England},
copyright = {The Author(s) 2024},
issn = {1354–0661},
journal = {European journal of international relations},
}

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@inproceedings{KooshaBehzad2025CAoR,
keywords = {Costs ; Earth ; Environmental monitoring ; Internet of
        Things ; Low earth orbit satellites ; Satellite constellations ;
        Satellites ; Space vehicles ; Telecommunications ; Timing},
language = {eng},
pages = {1–12},
publisher = {IEEE},
title = {Comprehensive Analysis of Recent LEO Satellite Constellations:
        Capabilities and Innovative Trends},
year = {2025},
author = {Koosha, Behzad and Madani, Pooria and Ardakani, Mansoor Dashti
        },
booktitle = {Proceedings – IEEE Aerospace Conference},
isbn = {9798350355970},
issn = {2996–2358},
}

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@inproceedings{LangleyAdam2017TQTP,
keywords = {Networks -- Network protocols -- Cross-layer protocols ;
        Networks -- Network protocols -- Network protocol design ; Networks
        -- Network protocols -- Transport protocols},
language = {eng},
pages = {183–196},
publisher = {ACM},
title = {The QUIC Transport Protocol: Design and Internet–Scale
        Deployment},
year = {2017},
author = {Langley, Adam and Riddoch, Alistair and Wilk, Alyssa and
        Vicente, Antonio and Krasic, Charles and Zhang, Dan and Yang, Fan and
        Kouranov, Fedor and Swett, Ian and Iyengar, Janardhan and Bailey, Jeff
        and Dorfman, Jeremy and Roskind, Jim and Kulik, Joanna and Westin,
        Patrik and Tenneti, Raman and Shade, Robbie and Hamilton, Ryan and
        Vasiliev, Victor and Chang, Wan–Teh and Shi, Zhongyi},

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address = {New York, NY, USA},
booktitle = {Proceedings of the Conference of the ACM Special Interest
    Group on Data Communication},
copyright = {2017 Owner/Author},
isbn = {1450346537},
}

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@article{DSRMethodology2007,
author = {Peffer, Ken and Tuunanen, Tuure and Rothenberger, Marcus and
    Chatterjee, S.},
year = {2007},
month = {01},
pages = {45-77},
title = {A design science research methodology for information systems
    research},
volume = {24},
journal = {Journal of Management Information Systems}
}

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@inproceedings{PolicastrA.J.2008_itP,
abstract = {PROTECT\texttrademark is an early-warning crisis management
    system for chemical attacks in high-threat infrastructures such as
    subways, transportation terminals, buildings, and airports. Current
    systems in NYC, Washington DC and Boston have been operated by the
    facility managers for the past several years. This paper presents some
    of the improvements made to the chemical detection system to (a)
    attain a very low detector false positive rate, (b) add new monitoring
    capabilities, and (c) reduce the cost of installation and maintenance
    . Ongoing work includes the addition of a wide range of detection
    devices and extension to biological, radiological, nuclear, and
    explosive detection.},
author = {Policastr, A.J.},
booktitle = {2008 IEEE Conference on Technologies for Homeland Security},
isbn = {1424419778},
keywords = {Automatic control ; Cameras ; Chemicals ; Costs ; Crisis
    management ; Detectors ; Laboratories ; Monitoring ; Protection ;
    Transportation},
language = {eng},
pages = {535-542},
publisher = {IEEE},
title = {"Advancements in the PROTECT\texttrademark Early-Warning Crisis
    Management System"},
year = {2008},
}

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@book{alma991972120506254,

abstract = {Evolving threats of chemical, biological, radiological and nuclear (CBRN) attack make it imperative to find ways to support global efforts against the proliferation of weapons of mass destruction (WMD) and terrorism. This book contains papers presented at the NATO Advanced Study Institute (ASI) on CBRN security culture, held in Yerevan, Armenia, in June 2014. The conference was timely in bringing together the various aspects of security culture in the different substantive areas from a CBRN perspective, focusing efforts to advance CBRN security culture in the everyday work of those who must deal with these ever present threats. The primary aim of the conference was to promote the concept of a CBRN security culture which recognizes synergy across the individual disciplines. As a first step in the practical application of this synergistic view, the conference introduced assessment methodologies that could be refined and customized to enhance the role of the human factor in CBRN security. This, we hope, will enable countries to fulfill their international obligations and implement United Nations Security Council Resolution 1540. The book reflects the presentations and deliberations of workshop participants and will be of interest to governments, international organizations, researchers and practitioners whose work involves the human dimension of the security of CBRN materials.},

author = {Thompson, Julia and Gahlaut, Seema},

address = {Amsterdam, [Netherlands]},

booktitle = {CBRN security culture in practice},

edition = {1st ed.},

isbn = {1-61499-534-6},

keywords = {Security International ; Terrorism -- Prevention ; Arms control},

language = {eng},

organization = {NATO Advanced Study Institute on CBRN Security Culture},

publisher = {IOS Press},

series = {NATO Science for Peace and Security Series. Sub-Series E: Human and Societal Dynamics, Volume 121},

title = {CBRN security culture in practice },

year = {2015 - 2015},

}